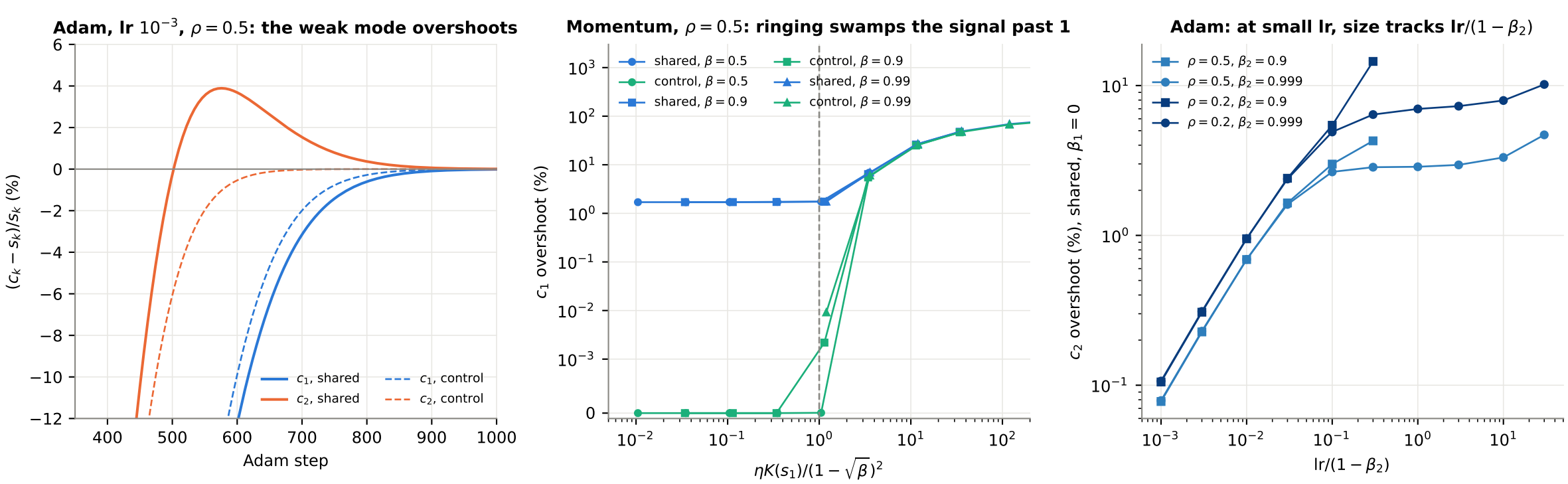
Adam, lr $10^{-3}$, $\rho = 0.5$: the weak mode overshoots
$(c_k - s_k)/s_k$ (%)
Adam step
$c_1$, shared
$c_2$, shared
$c_1$, control
$c_2$, control
Momentum, $\rho = 0.5$: ringing swamps the signal past 1
$c_1$ overshoot (%)
$\eta K(s_1)/(1-\sqrt{\beta})^2$
shared, $\beta = 0.5$
control, $\beta = 0.5$
shared, $\beta = 0.9$
control, $\beta = 0.9$
shared, $\beta = 0.99$
control, $\beta = 0.99$
Adam: at small lr, size tracks lr$/(1-\beta_2)$
$c_2$ overshoot (%), shared, $\beta_1 = 0$
lr$/(1-\beta_2)$
$\rho = 0.5$, $\beta_2 = 0.9$
$\rho = 0.5$, $\beta_2 = 0.999$
$\rho = 0.2$, $\beta_2 = 0.9$
$\rho = 0.2$, $\beta_2 = 0.999$